THE E_1 SPECTRAL SEQUENCE OF AN AUGMENTATION ON MODULES WITH E_1^{P,Q} = R^Q A_{P,*} K_P

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: functoriality in K (statement line 2404) rides on the cited Homology first-quadrant spectral sequence lemma, as recorded for card L46178; the row exactness uses the modules twin of card L46176's complex (lemma-augmentation-cech-complex-modules); no verification is omitted.

Editorial note. The label lemma-augmentation-spectral-sequence-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The ringed augmentation situation. Same chapter, Situation \label{situation-simplicial-site} and Remark \label{remark-augmentation-ringed}. Statement lines 2395-2397: K in D^+(O) for the sheaf of rings O on the total site; the pushforwards a_* and components a_{p,*}.
- The Cech complex of pushforwards for modules. Same chapter, \label{lemma-augmentation-cech-complex-modules}. Proof lines 2405-2413: the horizontal arrows of the double complex come from this lemma; its rows are exact in positive degrees and evaluate to a_* I^q in degree 0.
- Exactness of restriction and acyclicity of restricted injectives. Same chapter, \label{lemma-restriction-to-components-site} and \label{lemma-restriction-injective-to-component-limp}. Proof lines 2416-2422: restriction to C_p is exact, so I_p^bullet represents K_p; the I_p^q are totally acyclic on C_p.
- The homological-algebra machinery. Homology, Lemma \label{homology-lemma-double-complex-gives-resolution}, Derived Categories, Lemma \label{derived-lemma-leray-acyclicity}, Cohomology on Sites, Lemma \label{sites-cohomology-lemma-limp-acyclic}, Homology, Lemma \label{homology-lemma-first-quadrant-ss}. Proof lines 2413-2434: the total complex computes Ra_*K; the column cohomology is computed by Leray's acyclicity; the first-quadrant spectral sequence lemma concludes.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Prerequisite pointer invoked at proof line(s) 2429; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0133: sources/stacks-chow/tags.txt, source lines 1421–1421. Prerequisite pointer invoked at proof line(s) 2420; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0132: sources/stacks-chow/tags.txt, source lines 1420–1420. Prerequisite pointer invoked at proof line(s) 2433; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D7D: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2359–2359. Prerequisite pointer invoked at proof line(s) 2412; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D73: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1518–1518. Prerequisite pointer invoked at proof line(s) 2426; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 2422; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0731: sources/stacks-chow/tags.txt, source lines 9045–9045. Prerequisite pointer invoked at proof line(s) 2431; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation as above for any K in $D^+(\mathcal{O})$ there is a spectral sequence $(E_r, d_r)_{r\geq 0}$ in $Mod(\mathcal{O}_\mathcal{D})$ with*

$$E_1^{p,q} = R^q a_{p,*} K_p$$

converging to $R^{p+q}a_*K$. This spectral sequence is functorial in K.

Proof. Let $\mathcal{I}^\bullet$ be a bounded below complex of injective $\mathcal{O}$-modules representing K. Consider the double complex with terms

$$A^{p,q} = a_{p,*}\mathcal{I}_p^q$$

where the horizontal arrows come from Lemma 0D7D and the vertical arrows from the differentials of the complex $\mathcal{I}^\bullet$. The lemma says rows of the double complex are exact in positive degrees and evaluate to $a_*\mathcal{I}^q$ in degree 0. Thus the total complex associated to the double complex computes Ra_*K by Homology, Lemma 0133. On the other hand, since restriction to $\mathcal{C}_p$ is exact (Lemma 09WG) the complex $\mathcal{I}_p^\bullet$ represents K_p in $D(\mathcal{C}_p)$. The sheaves $\mathcal{I}_p^q$ are totally acyclic on $\mathcal{C}_p$ (Lemma 0D73). Hence the cohomology of the columns are the sheaves $R^q a_{p,*}K_p$ by Leray's acyclicity

lemma (Derived Categories, Lemma 015E) and Cohomology on Sites, Lemma 0731.
We conclude by applying Homology, Lemma 0132. □